Wine Dataset - Exploratory Data Analysis and Preprocessing

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from pandas.plotting import scatter_matrix
from scipy.stats import zscore
from sklearn.preprocessing import StandardScaler
```

1. Dataset Description & Understanding the Data Matrix

Dataset Chosen : Wine (UCI) - created by Stefan Aeberhard and M. Forina

Description : The Wine dataset contains chemical analysis of 178 wine samples from three cultivars in the same region of Italy. Each observation includes 13 continuous chemical measurements such as alcohol content, flavanoids, ash, etc. The class label indicates the wine cultivar.

1.1 Load Data & Display Dimensions.

In [4]:
```
columns = ["class","alcohol","malic_acid","ash","alcalinity_of_ash","magnesium","total_phenols","flavanoids","nonflavanoid_phe
```

In [6]:
```
df = pd.read_csv("wine.data", header = None, names = columns)
```

In [7]:
```
print(f"Dataset shape: {df.shape[0]} observations × {df.shape[1]} variables")
print(f"Number of observations (rows): {df.shape[0]}")
print(f"Number of variables (columns): {df.shape[1]} (1 class label + 13 features)")
print(f"\nClass distribution:\n{df['class'].value_counts().sort_index()}")
print("\nFirst 5 rows:")
df.head()
```

```
Dataset shape: 178 observations × 14 variables
Number of observations (rows): 178
Number of variables (columns): 14 (1 class label + 13 features)

Class distribution:
class
1    59
2    71
3    48
Name: count, dtype: int64

First 5 rows:
```

Out[7]:

	class	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	color_
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28	2.29	
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	2.81	
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	2.18	
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	1.82	

---

2. Exploratory Data Analysis (EDA)

2.1 Descriptive Statistics

In [11]:
```
quantitative_df = df.drop(columns = "class")
```

In [12]:
```
descriptive_stats = pd.DataFrame({
    "Min": quantitative_df.min(),
    "Max": quantitative_df.max(),
    "Range": quantitative_df.max() - quantitative_df.min(),
    "Mean": quantitative_df.mean(),
    "Median": quantitative_df.median(),
    "Variance": quantitative_df.var(),
    "Standard Deviation": quantitative_df.std()
})

descriptive_stats.round(3)
```

Out[12]:

	Min	Max	Range	Mean	Median	Variance	Standard Deviation
alcohol	11.03	14.83	3.80	13.001	13.050	0.659	0.812
malic_acid	0.74	5.80	5.06	2.336	1.865	1.248	1.117
ash	1.36	3.23	1.87	2.367	2.360	0.075	0.274
alcalinity_of_ash	10.60	30.00	19.40	19.495	19.500	11.153	3.340
magnesium	70.00	162.00	92.00	99.742	98.000	203.989	14.282
total_phenols	0.98	3.88	2.90	2.295	2.355	0.392	0.626
flavanoids	0.34	5.08	4.74	2.029	2.135	0.998	0.999
nonflavanoid_phenols	0.13	0.66	0.53	0.362	0.340	0.015	0.124
proanthocyanins	0.41	3.58	3.17	1.591	1.555	0.328	0.572
color_intensity	1.28	13.00	11.72	5.058	4.690	5.374	2.318
hue	0.48	1.71	1.23	0.957	0.965	0.052	0.229
od280_od315	1.27	4.00	2.73	2.612	2.780	0.504	0.710
proline	278.00	1680.00	1402.00	746.893	673.500	99166.717	314.907

Table 1 : Descriptive Statistics and Variable Characteristics of the Wine Dataset

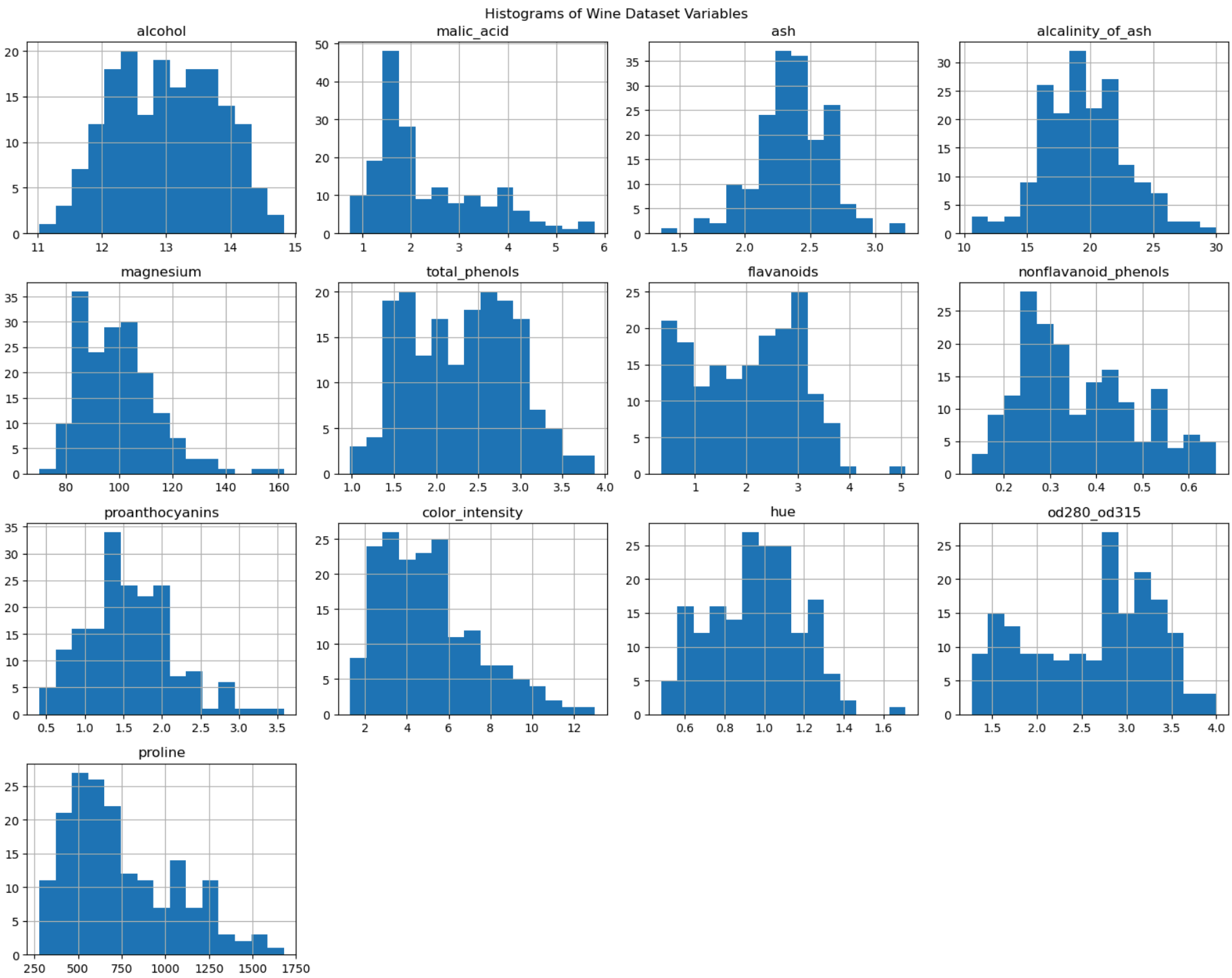
Variable	Type	Nature	Description	Minimum	Maximum	Range	Mean	Median	Variance	Standard Deviation
Class	Qualitative	Nominal	Wine cultivar (1, 2 or 3)							
Alcohol	Quantitative	Continuous	Alcohol content of the wine	11.03	14.83	3.80	13.001	13.050	0.659	0.812
Malic Acid	Quantitative	Continuous	Malic acid concentration	0.74	5.80	5.06	2.336	1.865	1.248	1.117
Ash	Quantitative	Continuous	Ash content of the wine	1.36	3.23	1.87	2.367	2.360	0.075	0.274
Alcalinity of Ash	Quantitative	Continuous	Alkalinity of ash	10.60	30.00	19.40	19.495	19.500	11.153	3.340
Magnesium	Quantitative	Discrete	Magnesium content	70.00	162.00	92.00	99.742	98.000	203.989	14.282
Total Phenols	Quantitative	Continuous	Total phenolic compounds	0.98	3.88	2.90	2.295	2.355	0.392	0.626
Flavanoids	Quantitative	Continuous	Flavanoid content	0.34	5.08	4.74	2.029	2.135	0.998	0.999
Non-flavanoid Phenols	Quantitative	Continuous	Non-flavanoid phenolic compounds	0.13	0.66	0.53	0.362	0.340	0.015	0.124
Proanthocyanins	Quantitative	Continuous	Proanthocyanin concentration	0.41	3.58	3.17	1.591	1.555	0.328	0.572
Color Intensity	Quantitative	Continuous	Intensity of wine color	1.28	13.00	11.72	5.058	4.690	5.374	2.318
Hue	Quantitative	Continuous	Hue of the wine	0.48	1.71	1.23	0.957	0.965	0.052	0.229
OD280/OD315	Quantitative	Continuous	Ratio of optical densities at 280nm and 315nm	1.27	4.00	2.73	2.612	2.780	0.504	0.710
Proline	Quantitative	Discrete	Proline amino acid content	278.00	1680.00	1402.00	746.893	673.500	99166.717	314.907

Highlights:

- **Scale differences are huge:** The variables are measured on very different scales; for example, proline has a much larger range than non-flavanoid phenols. This strongly motivates standardization before applying PCA.
- **Proline** and **magnesium** exhibit the largest variances, indicating the greatest raw spread due to measurement scale rather than intrinsic importance.
- For most variables, the mean and median are relatively close, suggesting limited skewness in their distributions.
- **Malic acid** and **color intensity** show noticeable mean–median differences, indicating possible right-skewness or the influence of high-value observations.

2.2 Histograms

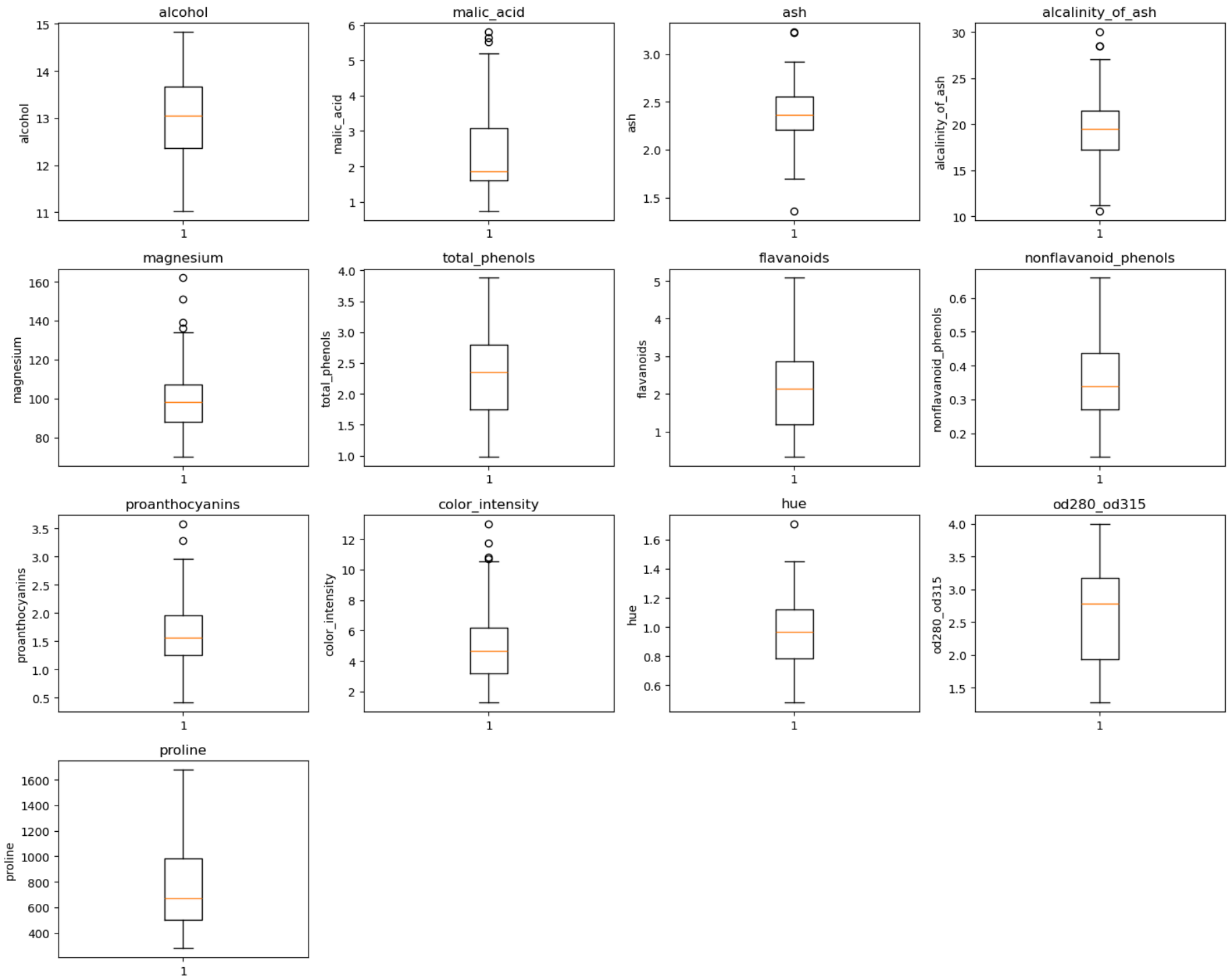
In [8]:
```
quantitative_df.hist(bins=15, figsize=(15, 12))
plt.suptitle("Histograms of Wine Dataset Variables")
plt.tight_layout()
plt.show()
```



Interpretation: Most variables appear unimodal with moderate spread, though several deviate from perfect symmetry. **Malic acid** and **color intensity** exhibit mild right-skew, indicating the presence of higher-value observations. **Proline** shows noticeable right-skew with a long tail toward larger values.

2.3 Boxplots

In [38]:
```
fig, axes = plt.subplots(4, 4, figsize=(15, 12))
axes = axes.flatten()
cols = ["alcohol","malic_acid","ash","alcalinity_of_ash","magnesium","total_phenols","flavanoids","nonflavanoid_phenols","proa


for ax, col in zip(axes, cols):
    ax.boxplot(df[col])
    ax.set_title(col)
    ax.set_ylabel(col)

# Remove empty subplot
fig.delaxes(axes[-1])
fig.delaxes(axes[-2])
fig.delaxes(axes[-3])

plt.tight_layout()
plt.show()
```

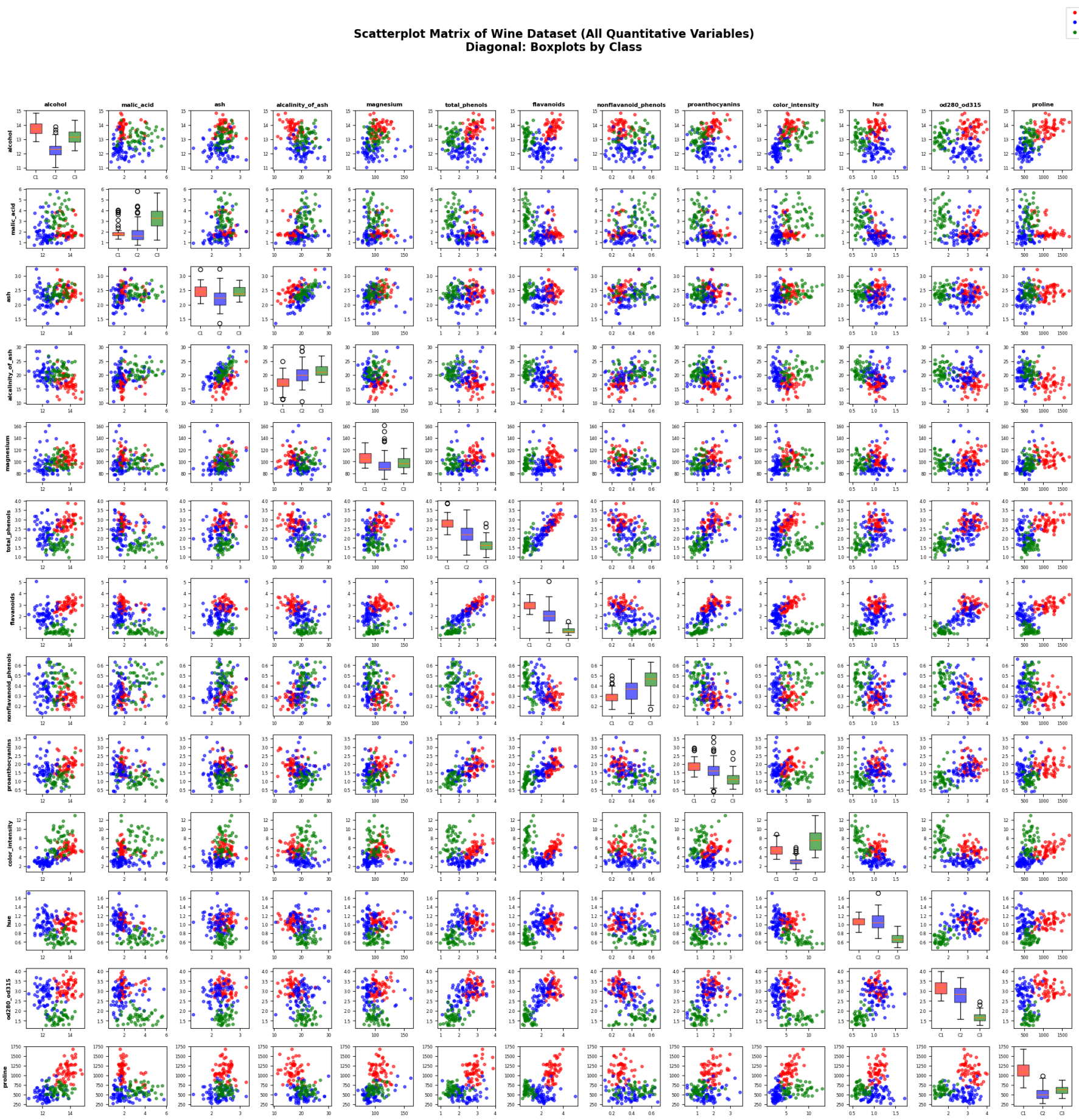


Interpretation: Several variables show observations beyond the whiskers in the boxplots, indicating the presence of potential outliers. **Malic acid**, **magnesium**, and **proline** exhibit particularly prominent high-value outliers, with additional outliers visible in variables such as **color intensity** and **alcalinity of ash**.

2.4 Scatterplot Matrix (coloured by class)

In [10]:
```
cols = ["alcohol","malic_acid","ash","alcalinity_of_ash","magnesium","total_phenols","flavanoids","nonflavanoid_phenols","proa

n = len(cols)
fig = plt.figure(figsize=(22, 22))
colors_map = {1: 'red', 2: 'blue', 3: 'green'}

for i in range(n):
    for j in range(n):
        ax = fig.add_subplot(n, n, i*n + j + 1)
        if i == j:
            data = [df.loc[df['class'] == cls, cols[i]]
                for cls in sorted(df['class'].unique())
            ]
            bp = ax.boxplot(data,patch_artist=True,widths=0.6,showfliers=True)

            for patch, cls in zip(bp['boxes'], sorted(df['class'].unique())):
                patch.set_facecolor(colors_map[cls])
                patch.set_alpha(0.6)

            ax.set_xticks(range(1, len(colors_map) + 1))
            ax.set_xticklabels([f"C{cls}" for cls in sorted(df['class'].unique())],fontsize=6)
        else:
            for cls in sorted(df['class'].unique()):
                mask = df['class'] == cls
                ax.scatter(df.loc[mask, cols[j]],df.loc[mask, cols[i]],c=colors_map[cls],alpha=0.6,s=15)
        if i == 0:
            ax.set_title(cols[j], fontsize=8, fontweight='bold')
        if j == 0:
            ax.set_ylabel(cols[i], fontsize=8, fontweight='bold')
        ax.tick_params(labelsize=6)

handles = [
    plt.Line2D([0], [0], marker='o', color='w',
               markerfacecolor=colors_map[cls], markersize=6,
               label=f'Class {cls}')
```

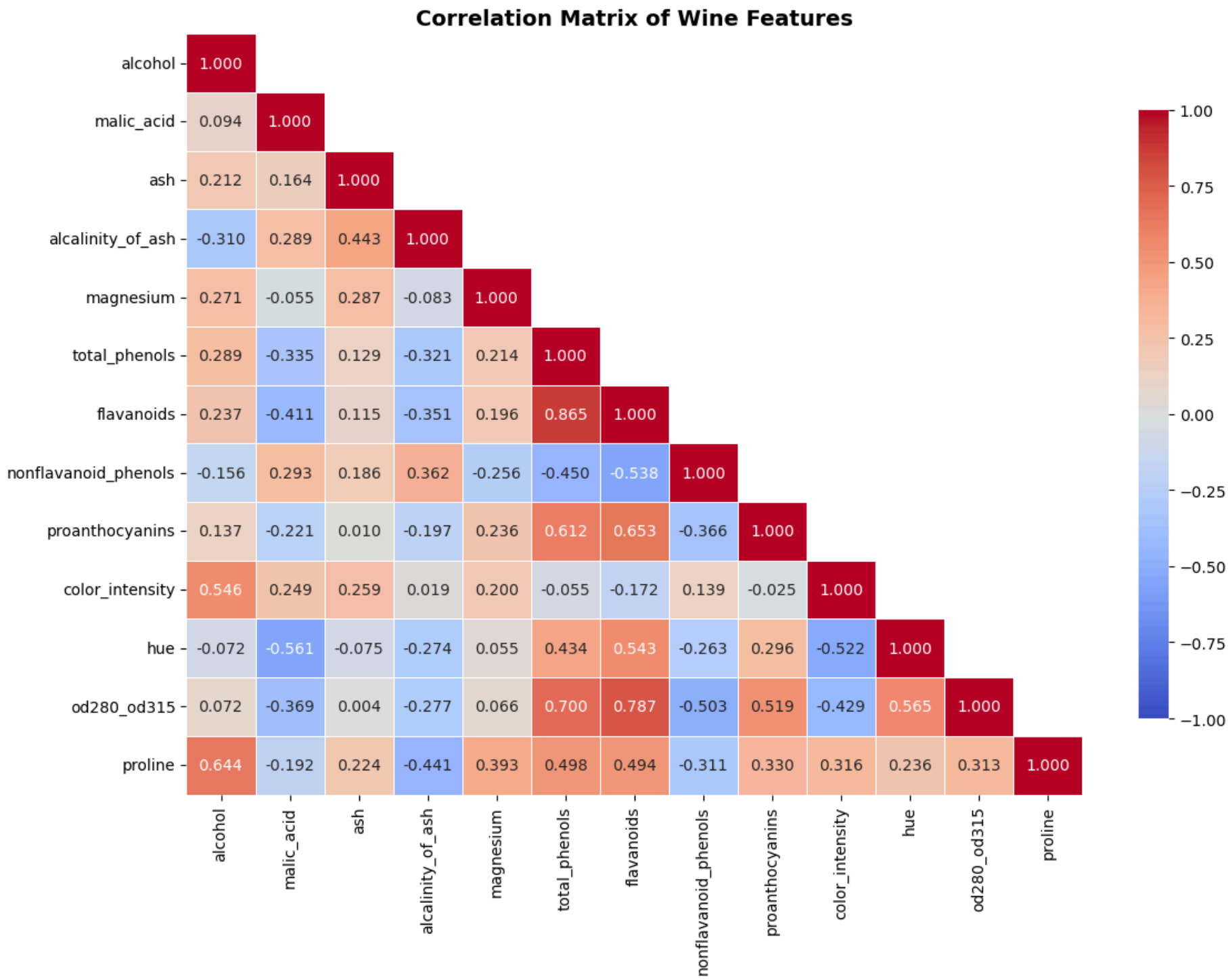
```
    for cls in sorted(df['class'].unique())
]

fig.legend(handles=handles, loc='upper right', fontsize=9)
fig.suptitle("Scatterplot Matrix of Wine Dataset (All Quantitative Variables)\nDiagonal: Boxplots by Class",fontsize=16,fontwe
plt.tight_layout(rect=[0, 0, 0.97, 0.95])
plt.show()
```



Interpretation: The three wine cultivars form visibly distinct groupings in several feature pairs, indicating structured multivariate variation and partial class separation. These patterns suggest the presence of correlated variables and dominant variance directions, which PCA can effectively summarize into a smaller number of uncorrelated components.

2.5 Correlation Matrix

In [17]:
```
corr_matrix = quantitative_df.corr()
fig, ax = plt.subplots(figsize=(12, 9))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool), k=1)
sns.heatmap(corr_matrix, mask=mask, annot=True, fmt='.3f', cmap='coolwarm', center=0,
            linewidths=0.5, ax=ax, vmin=-1, vmax=1, cbar_kws={'shrink': 0.8})
ax.set_title('Correlation Matrix of Wine Features', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



Interpretation: Strong positive correlations are observed among **Flavanoids**, **Total Phenols**, and **OD280/OD315**, indicating that these phenolic-related variables vary together. Several moderate negative correlations are also present, such as between **Malic Acid** and **Hue** and between **Nonflavanoid Phenols** and **Flavanoids**. Some variables, including Magnesium, exhibit weak correlations with many others.

---

3. Data Preprocessing

3.1 Missing Values

In [19]:
```
miss     = df.isnull().sum()
miss_pct = (miss / len(df) * 100)
miss_df  = pd.DataFrame({'Missing Count': miss, 'Missing %': miss_pct})
print(f"Total missing values in the dataset: {df.isnull().sum().sum()}\n")
miss_df
```

```
Total missing values in the dataset: 0
```

Out[19]:

	Missing Count	Missing %
class	0	0.0
alcohol	0	0.0
malic_acid	0	0.0
ash	0	0.0
alcalinity_of_ash	0	0.0
magnesium	0	0.0
total_phenols	0	0.0
flavanoids	0	0.0
nonflavanoid_phenols	0	0.0
proanthocyanins	0	0.0
color_intensity	0	0.0
hue	0	0.0
od280_od315	0	0.0
proline	0	0.0

The dataset does not contain missing values; therefore, no deletion or imputation is required.

3.2 Outlier Detection (Z-Score Method)

In [22]:
```
z_scores = np.abs(zscore(quantitative_df))
outlier_counts = (z_scores > 3).sum(axis=0)
outlier_df = pd.DataFrame({'Feature': cols,
                           'Outliers (|z| > 3)': outlier_counts})
total_outliers = (z_scores > 3).any(axis=1).sum()
print(f"Observations with |z| > 3 in at least one feature: {total_outliers} / {len(quantitative_df)}\n")
print(outlier_df.to_string(index=False))
```

```
Observations with |z| > 3 in at least one feature: 10 / 178

             Feature  Outliers (|z| > 3)
             alcohol                   0
          malic_acid                   1
                 ash                   3
   alcalinity_of_ash                   1
           magnesium                   2
       total_phenols                   0
          flavanoids                   1
nonflavanoid_phenols                   0
     proanthocyanins                   1
     color_intensity                   1
                 hue                   1
         od280_od315                   0
             proline                   0
```

Interpretation: A small number of observations are flagged as potential outliers based on the |z| > 3 criterion, affecting 10 out of 178 samples. These extreme values are limited in number and distributed across several variables, with no indication of systematic measurement error. Given that the data represent real chemical measurements and the outliers fall within plausible ranges, they are retained for analysis.

3.3 Standardisation

In [27]:
```
scaler = StandardScaler()

standardized_data = scaler.fit_transform(quantitative_df)

standardized_df = pd.DataFrame(
    standardized_data,
    columns=quantitative_df.columns
)

before_summary = quantitative_df.describe()

after_summary = standardized_df.describe()

print("Summary BEFORE standardization:\n")
before_summary.round(3)
```

```
Summary BEFORE standardization:
```

Out[27]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	color
count	178.000	178.000	178.000	178.000	178.000	178.000	178.000	178.000	178.000	
mean	13.001	2.336	2.367	19.495	99.742	2.295	2.029	0.362	1.591	
std	0.812	1.117	0.274	3.340	14.282	0.626	0.999	0.124	0.572	
min	11.030	0.740	1.360	10.600	70.000	0.980	0.340	0.130	0.410	
25%	12.362	1.602	2.210	17.200	88.000	1.742	1.205	0.270	1.250	
50%	13.050	1.865	2.360	19.500	98.000	2.355	2.135	0.340	1.555	
75%	13.678	3.082	2.558	21.500	107.000	2.800	2.875	0.438	1.950	
max	14.830	5.800	3.230	30.000	162.000	3.880	5.080	0.660	3.580	

In [28]:
```
print("\nSummary AFTER standardization:\n")
after_summary.round(3)
```

```
Summary AFTER standardization:
```

Out[28]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	color
count	178.000	178.000	178.000	178.000	178.000	178.000	178.000	178.000	178.000	
mean	-0.000	-0.000	-0.000	-0.000	-0.000	0.000	-0.000	0.000	-0.000	
std	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	1.003	
min	-2.434	-1.433	-3.679	-2.671	-2.088	-2.107	-1.696	-1.868	-2.069	
25%	-0.788	-0.659	-0.572	-0.689	-0.824	-0.885	-0.828	-0.740	-0.597	
50%	0.061	-0.423	-0.024	0.002	-0.122	0.096	0.106	-0.176	-0.063	
75%	0.836	0.670	0.698	0.602	0.510	0.809	0.849	0.610	0.629	
max	2.260	3.109	3.156	3.155	4.371	2.540	3.063	2.402	3.485	

Why standardise? In multivariate analysis, variables are often measured on very different scales. For example, proline values range in the hundreds, while nonflavanoid phenols take values less than one. In variance-based methods such as PCA, variables with larger numerical spread would dominate the analysis simply because they contribute more variance, not because they are more informative. Standardization, which centers variables to zero mean and scales them to unit variance, removes the effect of measurement units and ensures that PCA captures the underlying multivariate structure rather than differences in scale.

Compare summaries before and after standardization. Before standardization, the summary statistics are strongly influenced by the original measurement scales, causing large-scale variables to dominate variance. After standardization, all variables are centered and scaled to unit variance, removing scale effects and allowing meaningful comparison across variables. This ensures that multivariate techniques such as PCA capture genuine data structure rather than artifacts of differing units.

---

4. Preparation for PCA

4.1 Covariance Matrix of Standardised Data

Before Standardization :

In [23]:
```
cov_matrix_bfr_std = quantitative_df.cov()
cov_matrix_bfr_std
```

Out[23]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	p
alcohol	0.659062	0.085611	0.047115	-0.841093	3.139878	0.146887	0.192033	-0.015754	
malic_acid	0.085611	1.248015	0.050277	1.076332	-0.870780	-0.234338	-0.458630	0.040733	
ash	0.047115	0.050277	0.075265	0.406208	1.122937	0.022146	0.031535	0.006358	
alcalinity_of_ash	-0.841093	1.076332	0.406208	11.152686	-3.974760	-0.671149	-1.172083	0.150422	
magnesium	3.139878	-0.870780	1.122937	-3.974760	203.989335	1.916470	2.793087	-0.455563	
total_phenols	0.146887	-0.234338	0.022146	-0.671149	1.916470	0.391690	0.540470	-0.035045	
flavanoids	0.192033	-0.458630	0.031535	-1.172083	2.793087	0.540470	0.997719	-0.066867	
nonflavanoid_phenols	-0.015754	0.040733	0.006358	0.150422	-0.455563	-0.035045	-0.066867	0.015489	
proanthocyanins	0.063518	-0.141147	0.001516	-0.377176	1.932832	0.219373	0.373148	-0.026060	
color_intensity	1.028283	0.644838	0.164654	0.145024	6.620521	-0.079998	-0.399169	0.040121	
hue	-0.013313	-0.143326	-0.004682	-0.209118	0.180851	0.062039	0.124082	-0.007471	
od280_od315	0.041698	-0.292447	0.000762	-0.656234	0.669308	0.311021	0.558262	-0.044469	
proline	164.567185	-67.548867	19.319739	-463.355345	1769.158700	98.171057	155.447492	-12.203586	

After Standardization :

In [29]:
```
cov_matrix_aft_std = standardized_df.cov()
cov_matrix_aft_std
```

Out[29]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	pro
alcohol	1.005650	0.094930	0.212740	-0.311988	0.272328	0.290734	0.238153	-0.156810	
malic_acid	0.094930	1.005650	0.164972	0.290130	-0.054883	-0.337061	-0.413329	0.294632	
ash	0.212740	0.164972	1.005650	0.445872	0.288206	0.129708	0.115727	0.187283	
alcalinity_of_ash	-0.311988	0.290130	0.445872	1.005650	-0.083804	-0.322928	-0.353355	0.363966	
magnesium	0.272328	-0.054883	0.288206	-0.083804	1.005650	0.215613	0.196890	-0.257742	
total_phenols	0.290734	-0.337061	0.129708	-0.322928	0.215613	1.005650	0.869448	-0.452477	
flavanoids	0.238153	-0.413329	0.115727	-0.353355	0.196890	0.869448	1.005650	-0.540939	
nonflavanoid_phenols	-0.156810	0.294632	0.187283	0.363966	-0.257742	-0.452477	-0.540939	1.005650	
proanthocyanins	0.137470	-0.221993	0.009706	-0.198442	0.237776	0.615873	0.656379	-0.367912	
color_intensity	0.549451	0.250392	0.260350	0.018838	0.201080	-0.055448	-0.173353	0.139843	
hue	-0.072153	-0.564467	-0.075089	-0.275503	0.055711	0.436132	0.546549	-0.264123	
od280_od315	0.072752	-0.370794	0.003933	-0.278332	0.066377	0.703904	0.791641	-0.506113	
proline	0.647357	-0.193095	0.224890	-0.443086	0.395573	0.500929	0.496985	-0.313144	

Interpretation: Before standardization, the covariance matrix is heavily influenced by differences in measurement scale. Variables with large numerical ranges dominate both the variances and covariances, making it difficult to compare relationships across variable pairs. If PCA were applied at this stage, the resulting components would primarily reflect scale effects rather than the true multivariate structure.

---

5. Spectral Decomposition

5.1 Eigenvalues & Eigenvectors

Before Standardization :

In [33]:
```
eigvals_bfr, eigvecs_bfr = np.linalg.eigh(cov_matrix_bfr_std)

idx = np.argsort(eigvals_bfr)[::-1]
eigvals_bfr = eigvals_bfr[idx]
eigvecs_bfr = eigvecs_bfr[:, idx]

eig_df_bfr = pd.DataFrame({
    'PC'                 : [f'PC{i+1}' for i in range(len(eigvals_bfr))],
    'Eigenvalue'         : np.round(eigvals_bfr, 4),
    'Var Explained (%)'  : np.round(eigvals_bfr / eigvals_bfr.sum() * 100, 2),
    'Cumulative Var (%)' : np.round(np.cumsum(eigvals_bfr) / eigvals_bfr.sum() * 100, 2)
})
```

```python
print("Eigenvalues & Variance Explained (Before Standardization):\n")
print(eig_df_bfr.to_string(index=False))

evec_df_bfr = pd.DataFrame(eigvecs_bfr, index=cols, columns=[f'PC{i+1}' for i in range(len(eigvals_bfr))])

print("\nEigenvectors (Before Standardization):\n")
print(evec_df_bfr.round(4).to_string())
```

```
Eigenvalues & Variance Explained (Before Standardization):

  PC  Eigenvalue  Var Explained (%)  Cumulative Var (%)
 PC1  99201.7895              99.81               99.81
 PC2    172.5353               0.17               99.98
 PC3      9.4381               0.01               99.99
 PC4      4.9912               0.01              100.00
 PC5      1.2288               0.00              100.00
 PC6      0.8411               0.00              100.00
 PC7      0.2790               0.00              100.00
 PC8      0.1514               0.00              100.00
 PC9      0.1121               0.00              100.00
PC10      0.0717               0.00              100.00
PC11      0.0376               0.00              100.00
PC12      0.0211               0.00              100.00
PC13      0.0082               0.00              100.00

Eigenvectors (Before Standardization):

                        PC1     PC2     PC3     PC4     PC5     PC6     PC7     PC8     PC9    PC10    PC11    PC12    PC13
alcohol              0.0017 -0.0012 -0.0169  0.1414 -0.0203  0.1941 -0.9233 -0.2848  0.0866 -0.0022 -0.0150 -0.0157 -0.0080
malic_acid          -0.0007 -0.0022 -0.1220  0.1604  0.6129  0.7425  0.1501  0.0647  0.0157 -0.0185 -0.0232  0.0673  0.0111
ash                  0.0002 -0.0046 -0.0520 -0.0098 -0.0202  0.0418 -0.0450  0.1493  0.0736 -0.0868  0.9540 -0.1321  0.1737
alcalinity_of_ash   -0.0047 -0.0265 -0.9386 -0.3310 -0.0644 -0.0241 -0.0315 -0.0152  0.0020  0.0036 -0.0528  0.0054 -0.0019
magnesium            0.0179 -0.9993  0.0298 -0.0054  0.0061 -0.0019 -0.0018  0.0036 -0.0020 -0.0000 -0.0030  0.0006 -0.0023
total_phenols        0.0010 -0.0009  0.0405 -0.0746 -0.3152  0.2787  0.0202  0.1772  0.2557  0.8472  0.0088  0.0039  0.0267
flavanoids           0.0016  0.0001  0.0854 -0.1691 -0.5248  0.4336  0.0389  0.2481  0.3783 -0.5201 -0.1332 -0.0375 -0.0696
nonflavanoid_phenols -0.0001  0.0014 -0.0135  0.0108  0.0296 -0.0220  0.0047 -0.0065  0.0368  0.0377  0.1992  0.1476 -0.9665
proanthocyanins      0.0006 -0.0050  0.0247 -0.0501 -0.2512  0.2419  0.3098 -0.8704 -0.0515  0.0097  0.1356 -0.0131  0.0176
color_intensity      0.0023 -0.0151 -0.2914  0.8789 -0.3317  0.0027  0.1128  0.0813 -0.0990 -0.0231 -0.0098  0.0504  0.0046
hue                  0.0002  0.0008  0.0260 -0.0600 -0.0515 -0.0238 -0.0308  0.0030  0.0331 -0.0385  0.0975  0.9756  0.1666
od280_od315          0.0007  0.0035  0.0703 -0.1782 -0.2606  0.2889 -0.1020  0.1867 -0.8737  0.0170  0.0285  0.0116 -0.0442
proline              0.9998  0.0178 -0.0045 -0.0031  0.0023 -0.0012  0.0011 -0.0000 -0.0001  0.0000 -0.0002 -0.0001 -0.0000
```

After Standardization :

In [31]:
```python
eigvals_aft, eigvecs_aft = np.linalg.eigh(cov_matrix_aft_std)

idx = np.argsort(eigvals_aft)[::-1]
eigvals_aft = eigvals_aft[idx]
eigvecs_aft = eigvecs_aft[:, idx]

eig_df_aft = pd.DataFrame({
    'PC'                 : [f'PC{i+1}' for i in range(len(eigvals_aft))],
    'Eigenvalue'         : np.round(eigvals_aft, 4),
    'Var Explained (%)'  : np.round(eigvals_aft / eigvals_aft.sum() * 100, 2),
    'Cumulative Var (%)' : np.round(np.cumsum(eigvals_aft) / eigvals_aft.sum() * 100, 2)
})

print("Eigenvalues & Variance Explained (After Standardization):\n")
print(eig_df_aft.to_string(index=False))

evec_df_aft = pd.DataFrame(eigvecs_aft, index=cols, columns=[f'PC{i+1}' for i in range(len(eigvals_aft))])

print("\nEigenvectors (After Standardization):\n")
print(evec_df_aft.round(4).to_string())
```

```
Eigenvalues & Variance Explained (After Standardization):

  PC  Eigenvalue  Var Explained (%)  Cumulative Var (%)
 PC1      4.7324              36.20               36.20
 PC2      2.5111              19.21               55.41
 PC3      1.4542              11.12               66.53
 PC4      0.9242               7.07               73.60
 PC5      0.8580               6.56               80.16
 PC6      0.6453               4.94               85.10
 PC7      0.5541               4.24               89.34
 PC8      0.3505               2.68               92.02
 PC9      0.2905               2.22               94.24
PC10      0.2523               1.93               96.17
PC11      0.2271               1.74               97.91
PC12      0.1697               1.30               99.20
PC13      0.1040               0.80              100.00

Eigenvectors (After Standardization):

                        PC1     PC2     PC3     PC4     PC5     PC6     PC7     PC8     PC9    PC10    PC11    PC12    PC13
alcohol             -0.1443  0.4837  0.2074 -0.0179 -0.2657 -0.2135  0.0564 -0.3961  0.5086  0.2116  0.2259  0.2663  0.0150
malic_acid           0.2452  0.2249 -0.0890  0.5369  0.0352 -0.5368 -0.4205 -0.0658 -0.0753 -0.3091 -0.0765 -0.1217  0.0260
ash                  0.0021  0.3161 -0.6262 -0.2142 -0.1430 -0.1545  0.1492  0.1703 -0.3077 -0.0271  0.4987  0.0496 -0.1412
alcalinity_of_ash    0.2393 -0.0106 -0.6121  0.0609  0.0661  0.1008  0.2870 -0.4280  0.2004  0.0528 -0.4793  0.0557  0.0917
magnesium           -0.1420  0.2996 -0.1308 -0.3518  0.7270 -0.0381 -0.3229  0.1564  0.2714  0.0679 -0.0713 -0.0622  0.0568
total_phenols       -0.3947  0.0650 -0.1462  0.1981 -0.1493  0.0841  0.0279  0.4059  0.2860 -0.3201 -0.3043  0.3039 -0.4639
flavanoids          -0.4229 -0.0034 -0.1507  0.1523 -0.1090  0.0189  0.0607  0.1872  0.0496 -0.1632  0.0257  0.0429  0.8323
nonflavanoid_phenols  0.2985  0.0288 -0.1704 -0.2033 -0.5007  0.2586 -0.5954  0.2333  0.1955  0.2155 -0.1169 -0.0424  0.1140
proanthocyanins     -0.3134  0.0393 -0.1495  0.3991  0.1369  0.5338 -0.3721 -0.3682 -0.2091  0.1342  0.2374  0.0956 -0.1169
color_intensity      0.0886  0.5300  0.1373  0.0659 -0.0764  0.4186  0.2277  0.0338  0.0562 -0.2908 -0.0318 -0.6042 -0.0120
hue                 -0.2967 -0.2792 -0.0852 -0.4278 -0.1736 -0.1060 -0.2321 -0.4366  0.0858 -0.5224  0.0482 -0.2592 -0.0899
od280_od315         -0.3762 -0.1645 -0.1660  0.1841 -0.1012 -0.2659  0.0448  0.0781  0.1372  0.5237 -0.0464 -0.6010 -0.1567
proline             -0.2868  0.3649  0.1267 -0.2321 -0.1579 -0.1197 -0.0768 -0.1200 -0.5758  0.1621 -0.5393  0.0794  0.0144
```

---

6. Interpretation & Summary

Large vs Small Eigenvalues

A **large eigenvalue** means that the corresponding principal component captures a large fraction of the total variance in the data. Conversely, a **small eigenvalue** indicates that very little variability lies along that direction i.e., that component is nearly redundant.
From the decomposition, **PC1 alone accounts for ~36 %** of the total variance, and the first 3 PCs together explain **~60 %**. The last few PCs (PC10 –PC13) each contribute less than 2 %, meaning those directions can be safely discarded in a dimensionality-reduction step.

Orthogonality of Eigenvectors

The eigenvectors of a symmetric matrix (like the covariance matrix) are **mutually orthogonal**: their dot products are zero. Geometrically this means the new coordinate axes (principal components) are **perpendicular** to each other, so the components are **uncorrelated by construction**. This orthogonality is what makes PCA such a clean decomposition i.e., each PC captures a unique, non-overlapping slice of variance.

Eigenvalues as Variability

Each eigenvalue λ_i quantifies the variance of the data when projected onto the i-th eigenvector. Summing all eigenvalues gives the **total variance** of the dataset (trace of the covariance matrix for standardised data). The ratio $\lambda_i / \Sigma\lambda$ gives the proportion of total variance explained by PC_i.

Eigenvectors as New Directions

Each eigenvector is a **unit vector** in the original 13-dimensional quantitative feature space. Its components tell us how much each original feature contributes to that principal component.

Dimensionality Reduction via PCA

PCA reduces dimensionality by keeping only the top-k principal components (those with the largest eigenvalues). The projection of each observation onto these k directions retains the **maximum possible variance** among all k-dimensional subspaces. If the first k PCs already capture, say, 95 % of total variance, the remaining 13 − k dimensions add very little information and can be discarded, thus shrinking the feature space while preserving the essential structure of the data.